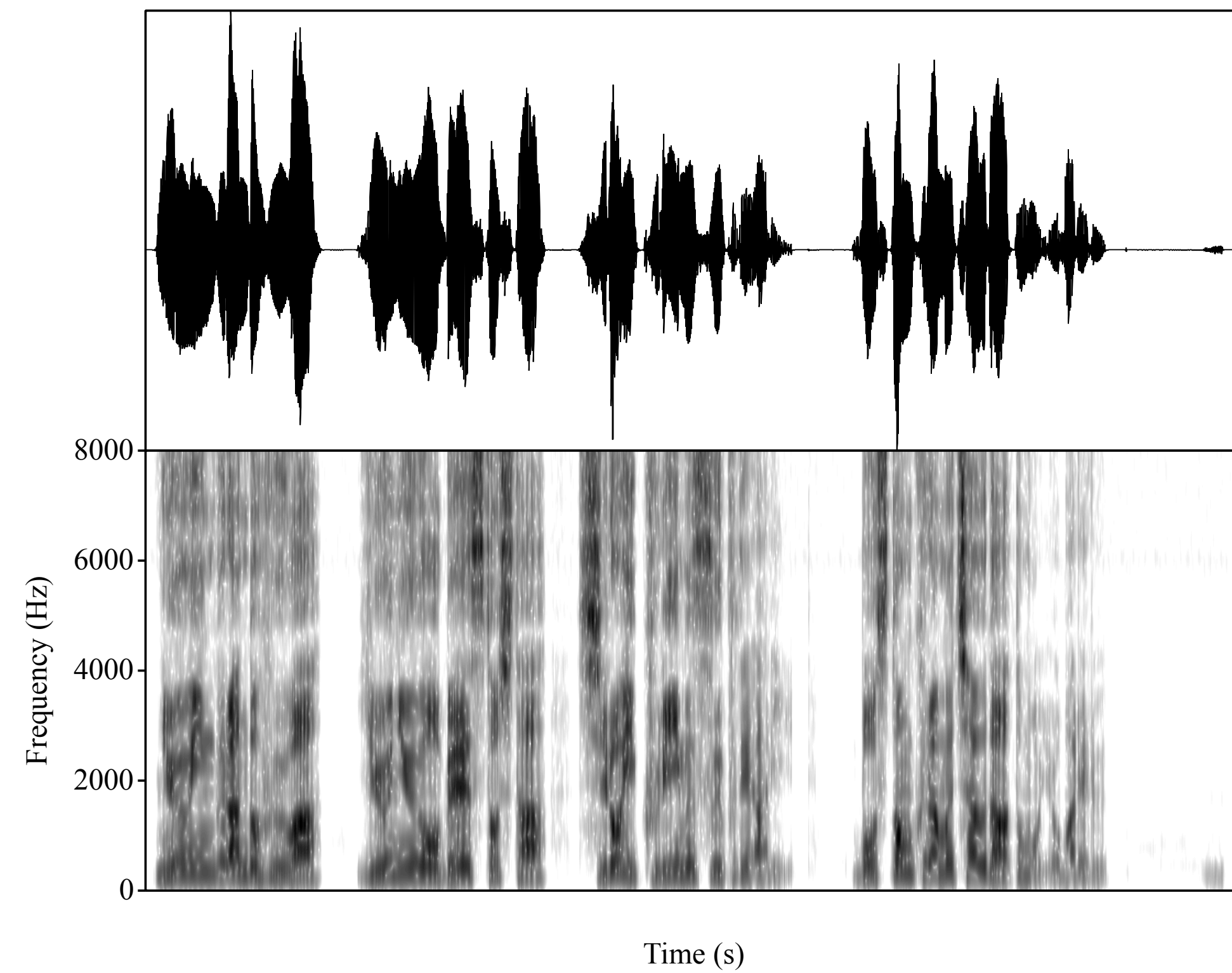
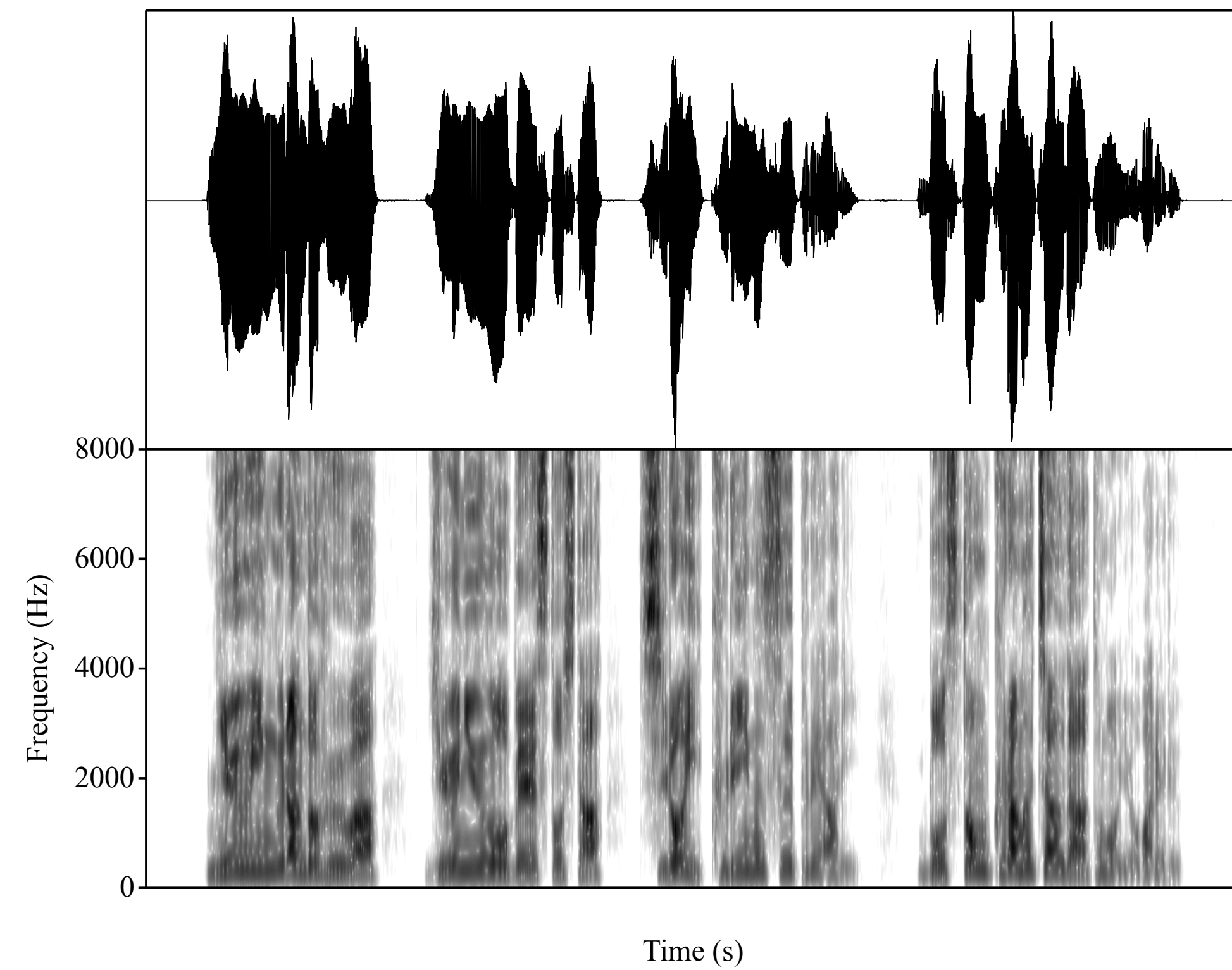


Making sense of AI-generated speech

Challenges and opportunities for phonetic and speech science research

James Tanner 2025/10/27 Waseda University





British 999 call handler's voice cloned by Russian network using AI

A BBC Verify investigation has revealed that the identities of British public sector workers have been cloned using AI by a Russian-linked disinformation campaign.

朝日新聞 > 連載 > 新世AI > 記事

第44回

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🔒 有料記事

宮坂麻子 2024年12月24日 11時30分

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 - Often require 'audio watermarking' of generated speech
 - Detection models are also 'black boxes' with limited explainability

ASVspoof 5: Design, Collection and Validation of Resources for Spoofing, Deepfake, and Adversarial Attack Detection Using Crowdsourced Speech

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Detect whether an audio clip was created using ElevenLabs

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A Review of Modern Audio Deepfake Detection Methods: Challenges and Future Directions

by Zaynab Almutairi^{1,*}  and Hebah Elgibreen^{1,2} 

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Do we need an alternative approach?

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 - *Feature explainability: how do particular features relate to the output?*

AI explainability

- While we can describe an AI speech synthesis model *mechanistically*, it tells us little about how a model *actually* learns and produces language
- In other words, we *lack featural explainability* for AI-generated speech models

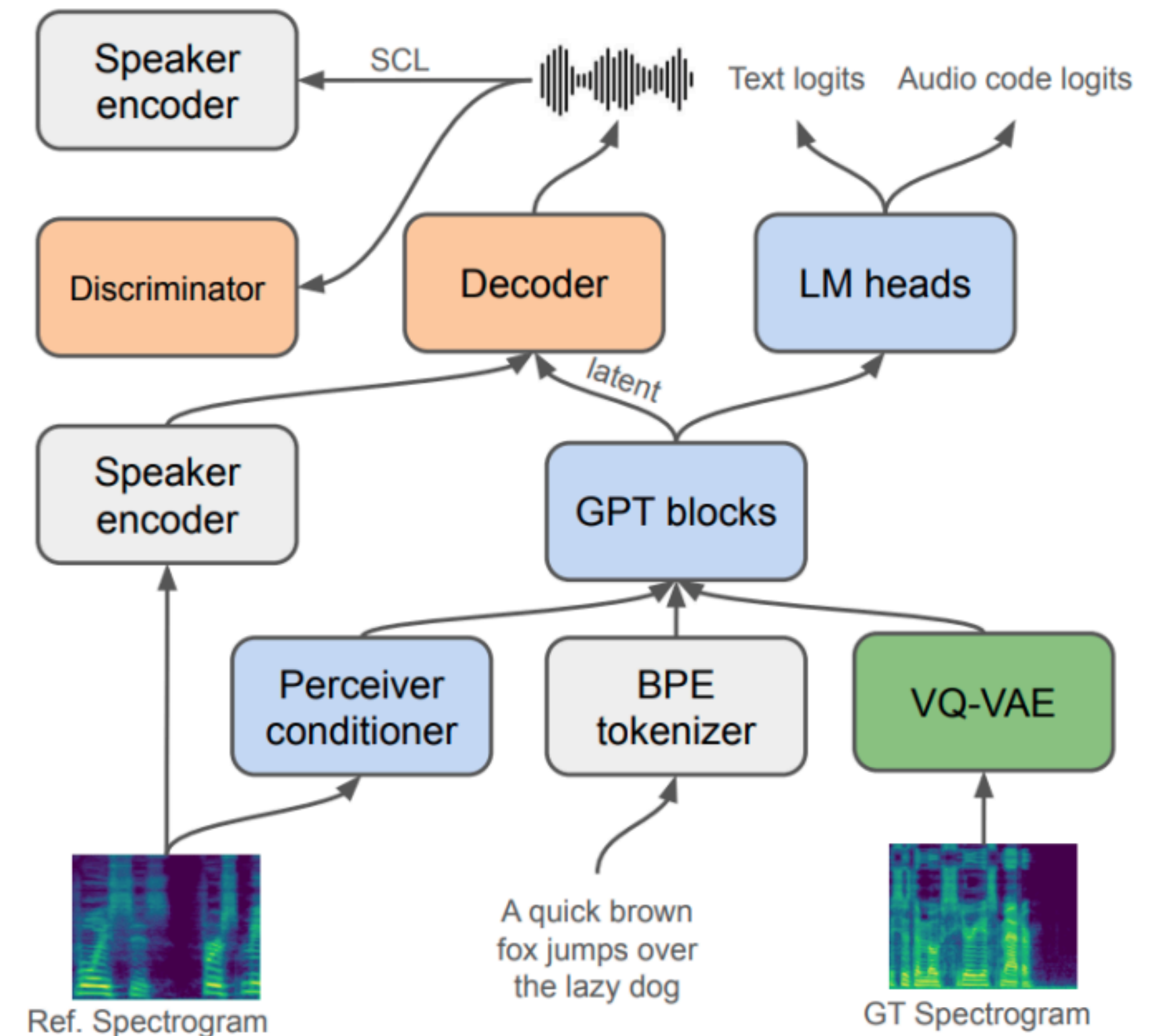


Figure 1: *XTTS training architecture overview.*

Introduction

- Misuse of AI-generated speech poses risks for security, privacy, and personal identity
- Yet we still understand little about AI-generated speech itself:
 - *What are AI speech models capturing when cloning a speaker?*
 - *In what ways is AI-generated speech similar/different to human speech?*
 - *How do AI models learn language-specific vs speaker-specific properties?*

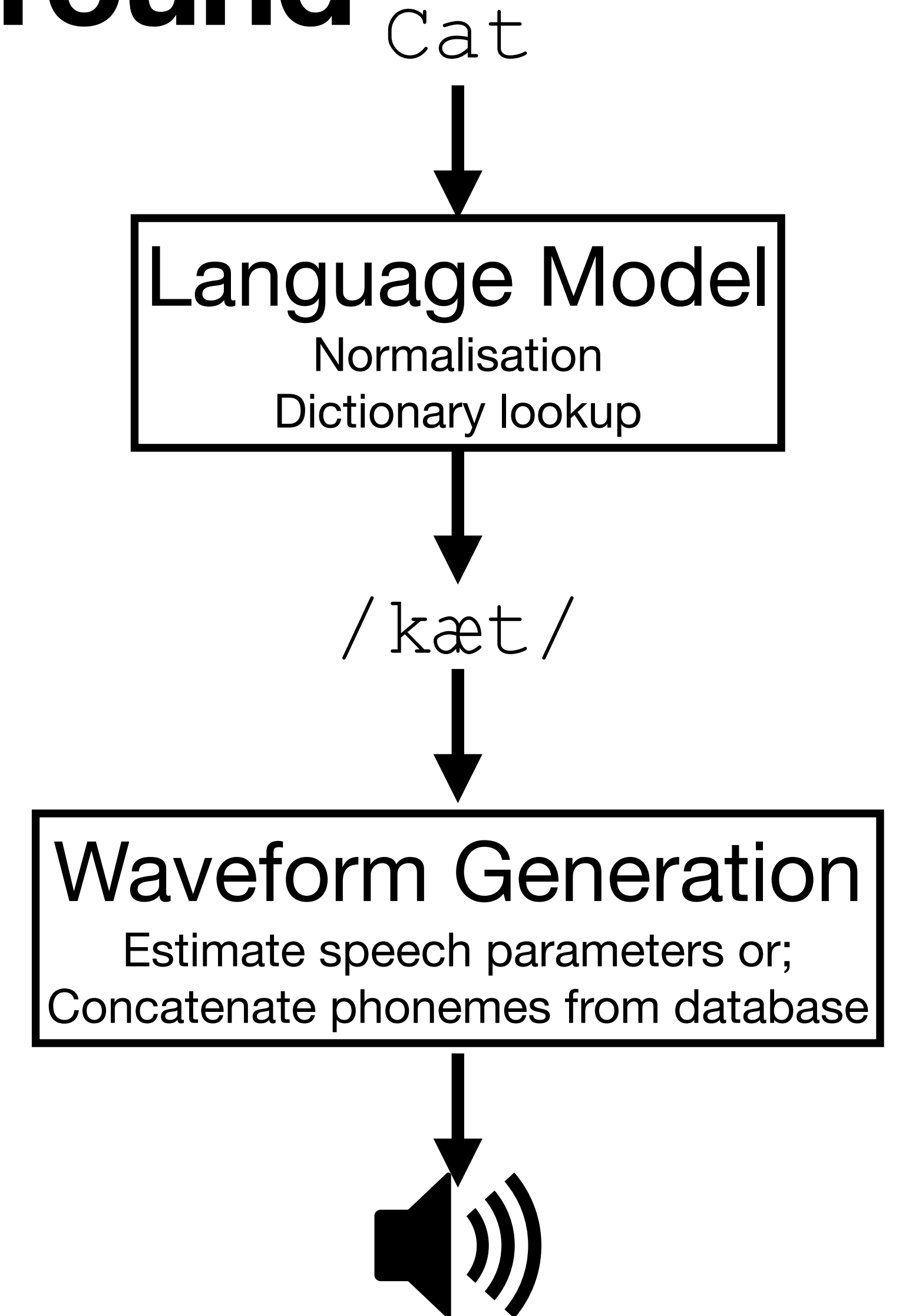
This talk

- Introduction to AI speech synthesis and ‘voice cloning’
- Can phonetics help?
- Case study: Phonetic variation in English & Japanese AI-generated stops
- Future directions

What is AI-generated speech?

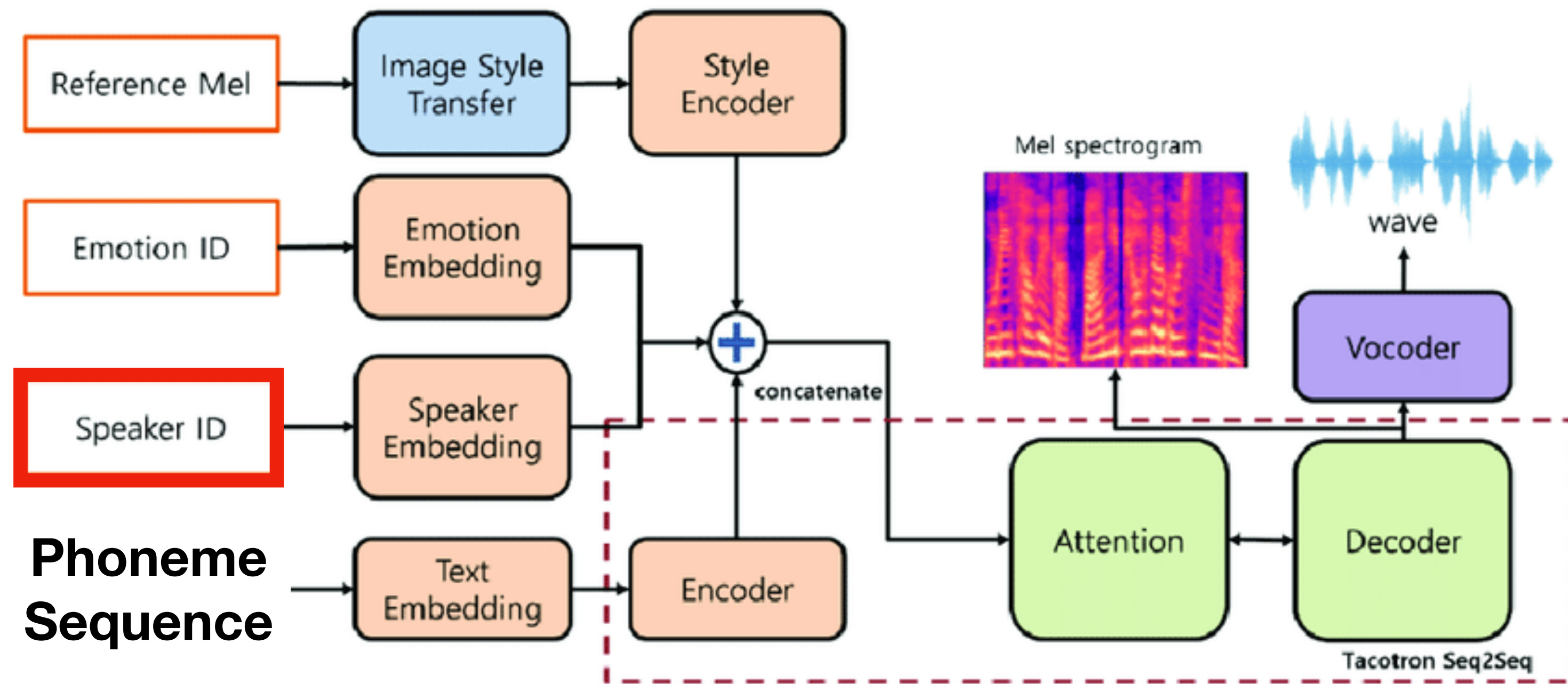
TTS speech synthesis: background

- Generate (speech) audio waveform based on text string
- Historically (until ~2017):
- Text-processing ('front-end')
 - *Language model*: either set of rules (usually) or probabilistic
 - Output: phoneme string
- Generate waveform ('back-end')
 - Parametric synthesis: estimate acoustic-phonetic parameters and generate
 - Concatenative/'unit selection' synthesis: create database of phonemes (from real speech) place them in order
- Large (phonetically-balanced) corpora *for each speaker* -> \$\$\$



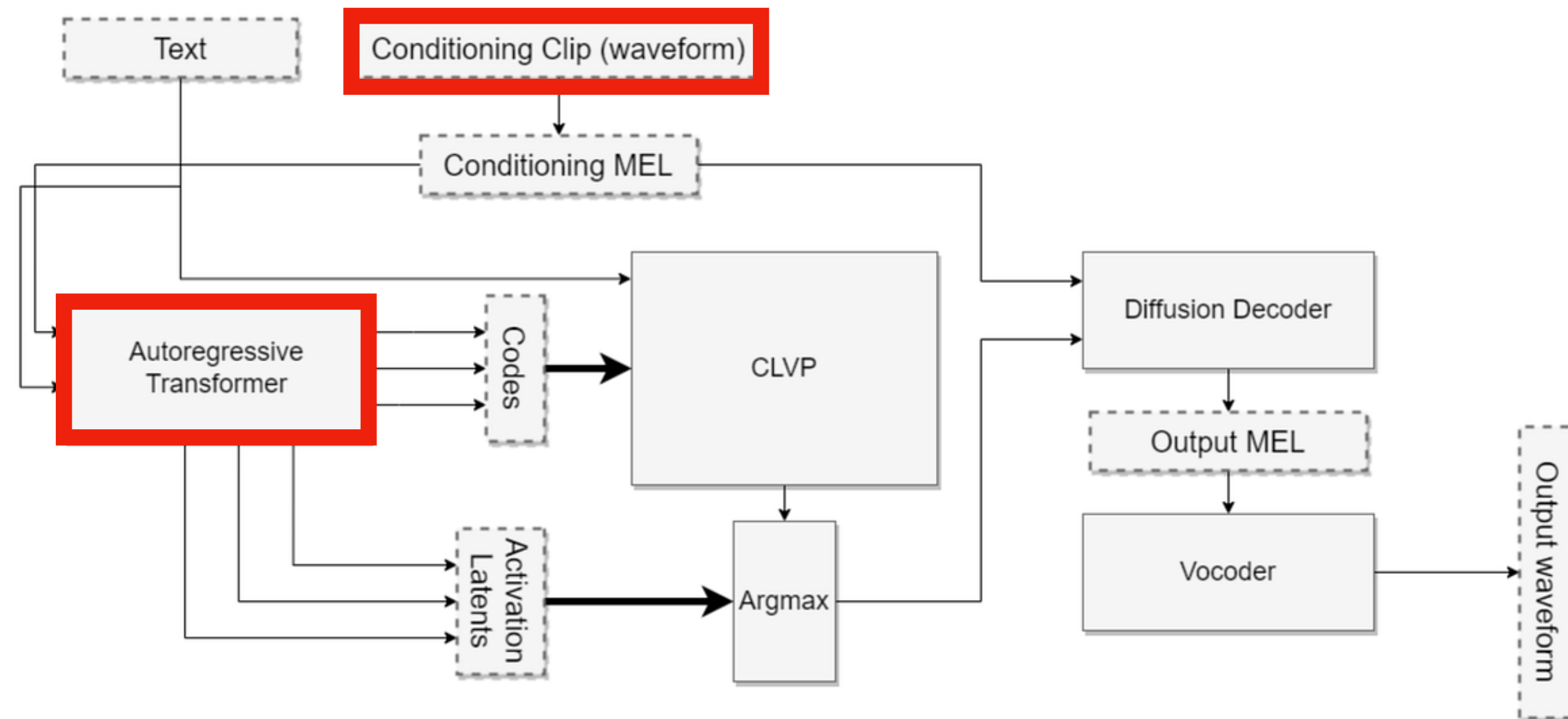
Tokuda *et al.* (2013), King (2014)

2017: Tacotron



- Neural network to estimate spectrogram from phoneme sequence
- Speaker embedding:
 - High-dimensional representation of speaker similarities/differences
 - Can only produce speaker it has been trained on
 - Makes it possible to *leverage* information from ‘similar’ speakers
 - **Massively** reduces the amount of speech data needed for a given speaker

2022: Tortoise



- Replace text processing with Pretrained Transformer ('GPT')
 - Learns the mapping between orthographic text and waveform directly
- Model doesn't need to be trained in advance for a particular speaker
 - Reference audio for target speaker used to *condition* ('prompt') GPT to produce audio in speaker's voice
 - "Voice cloning" with <1 minute of speech
 - Enables cloning speaker into other languages

2023-present

- Models refining Tortoise's GPT-based approach for stability, generalisability
 - VITS, X-TTS, Fish-Speech, Chatterbox etc
- Many now widely-available and **open source**
- Anyone (with a GPU and basic Python knowledge) can use models to generate speech of *any speaker for free*

2023-present

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```
model = ChatterboxMultilingualTTS.from_local("/path/to/model")
```

```
input_wav = "/path/to/speaker01.wav"
```

```
output_wav = "nisemono.wav"
```

```
text = "ニューイングランド風は、牛乳をベースとした、白い  
クリームスープであり、ボストンクラムチャウダーとも呼ばれる。"
```

```
wav = model.generate(text, input_wav, language = "ja")  
save(output_wav, wav)
```

Can phonetics help?

What do we know about AI generated speech?

- Speech synthesis performance is evaluated *impressionistically*
 - Preference (A/B) tests, rating tests
 - Speech recognition performance

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 - TTS models produce different spectra for a single speaker's obstruents

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- Previous research (on older models):
 - Listeners are better at distinguishing context-sensitive variation in newer TTS systems
 - TTS models produce different spectra for a single speaker's obstruents
- Despite widespread AI speech usage, **very little** work has examined what generated speech is like *phonetically*

Zellou *et al.* (2021), Taylor & Richmond (2021), Pandey *et al.* (2022),
Kirkland *et al.* (2023), Bhyravajjula *et al.* (2025)

What do we know about AI generated speech?

- However, decades of linguistic research has:
 - Shown the ways in which speech can systematically vary
 - Vowel duration: high -> low
 - Stop duration: voiceless -> voiced
 - Vowel quality: position of vowels in F1xF2 space

What do we know about AI generated speech?

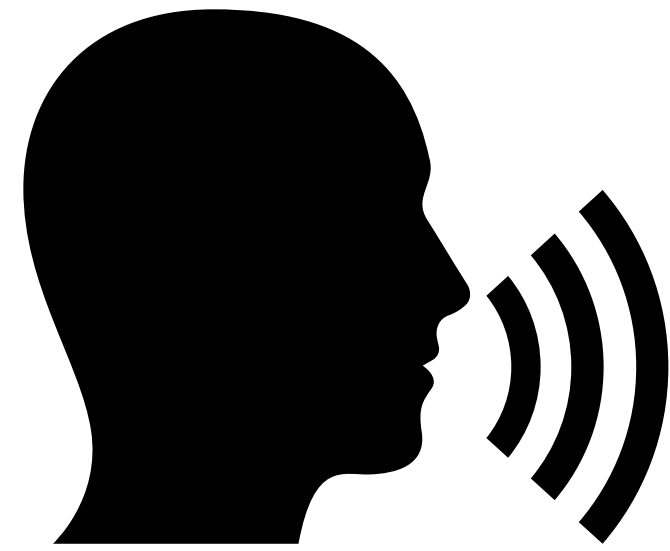
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 - Vowel duration: high -> low
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 - Vowel quality: position of vowels in F1xF2 space
 - Developed methods for the *interpretable analysis* of speech and speaker differences
 - Acoustic-phonetic analysis
 - Forensic methods

Peterson & Barney (1952), Hillenbrand *et al.* (1995), Boersma & Weenik (1992-), Abramson (2017), Morrison & Enzinger (2019)

A phonetic approach to the study of AI speech

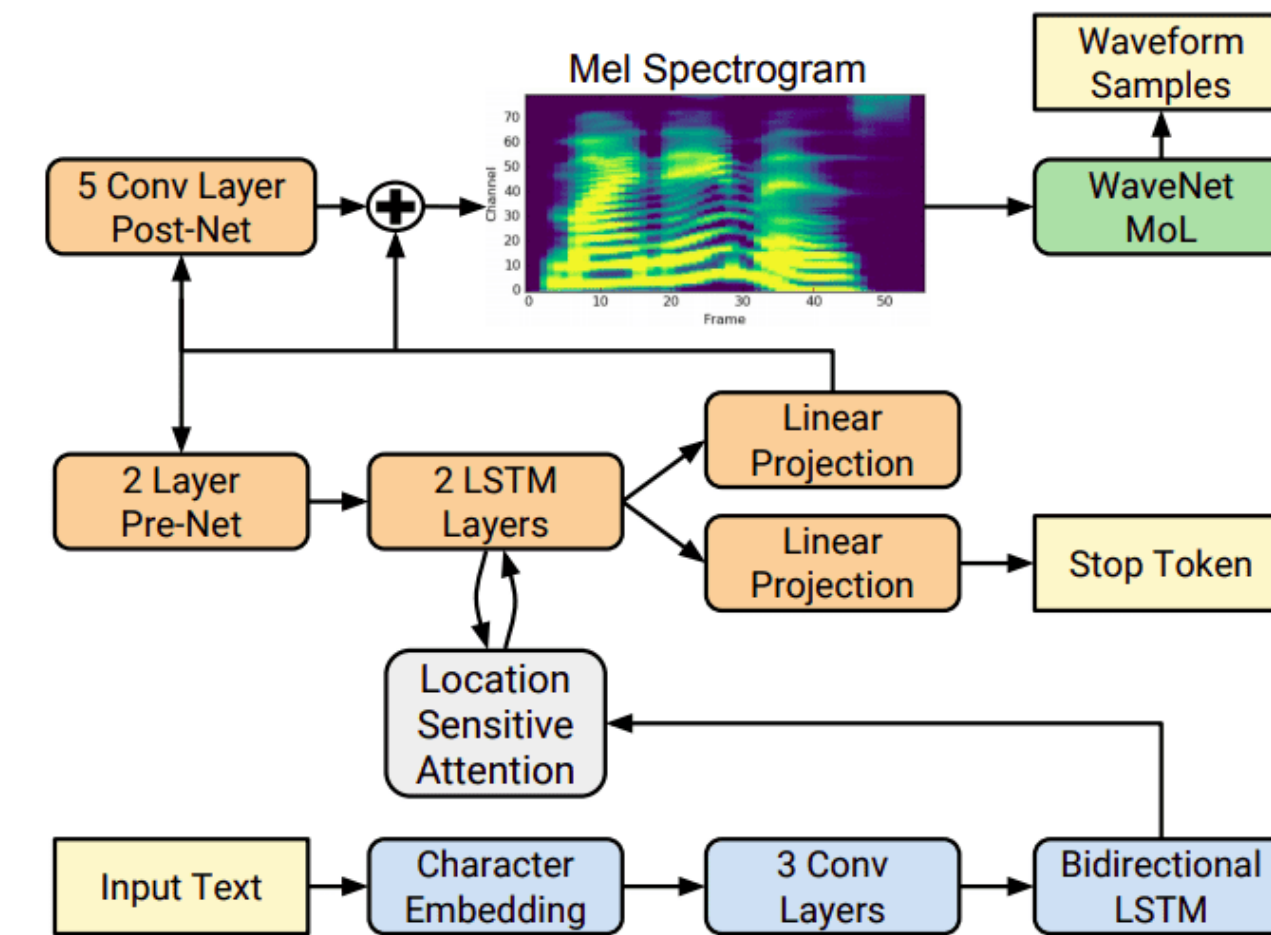
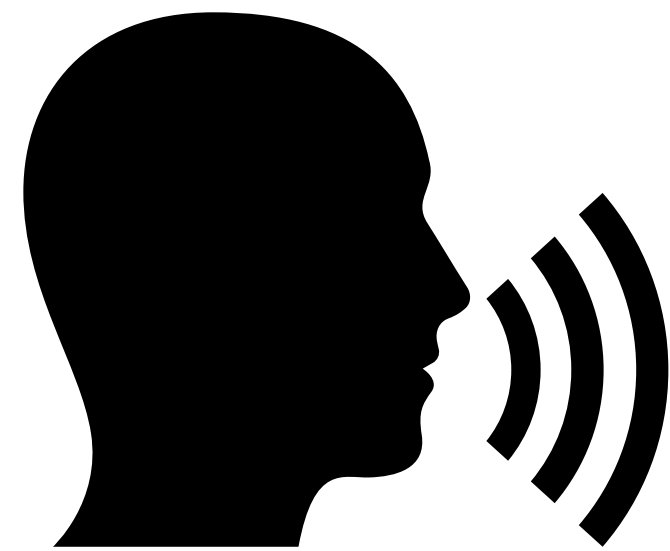
- Allow us to ask phonetically-motivated questions about AI-generated speech
 - *Do AI-generated vowels pattern the same as human vowels?*
 - *What do AI-generated vowel spaces look like?*
 - *Do AI models produce different /t/ realisations for English and Japanese?*
 - *What properties of a speaker's acoustic profile are present in the voice clone?*

What would an AI speech study look like?



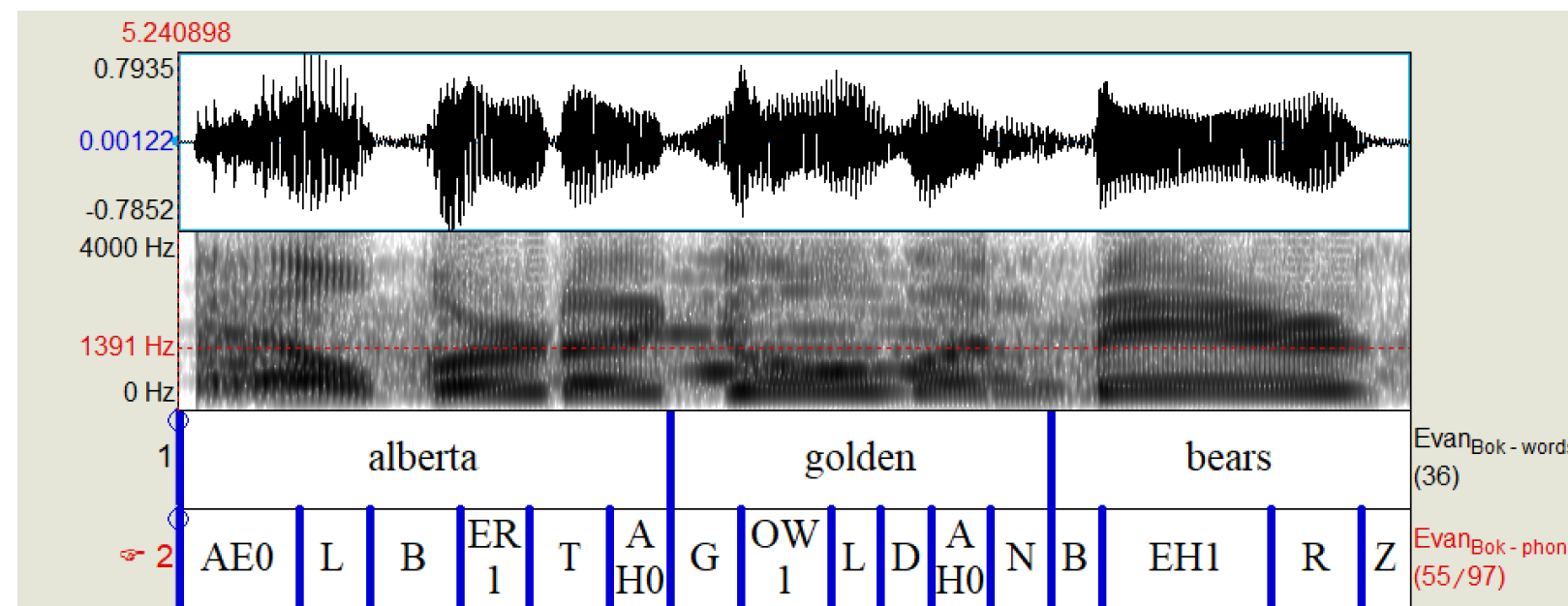
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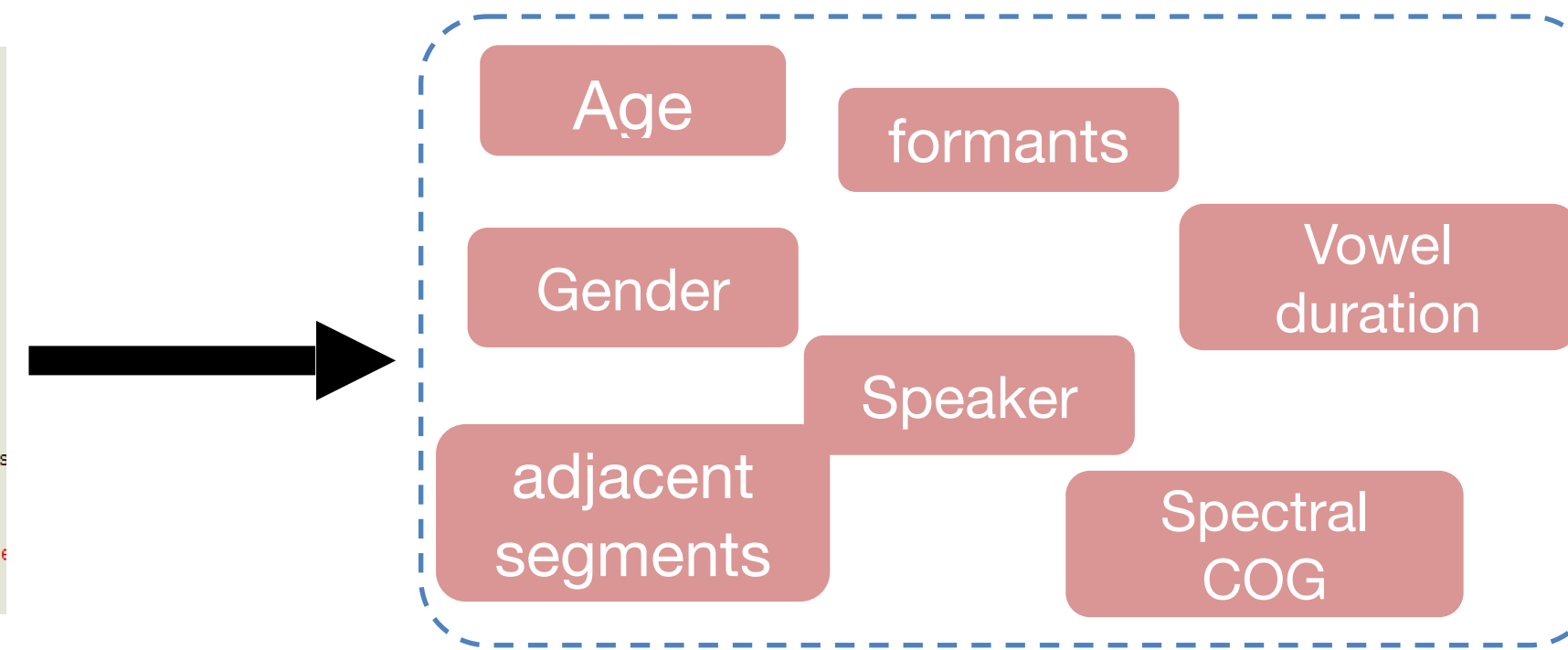
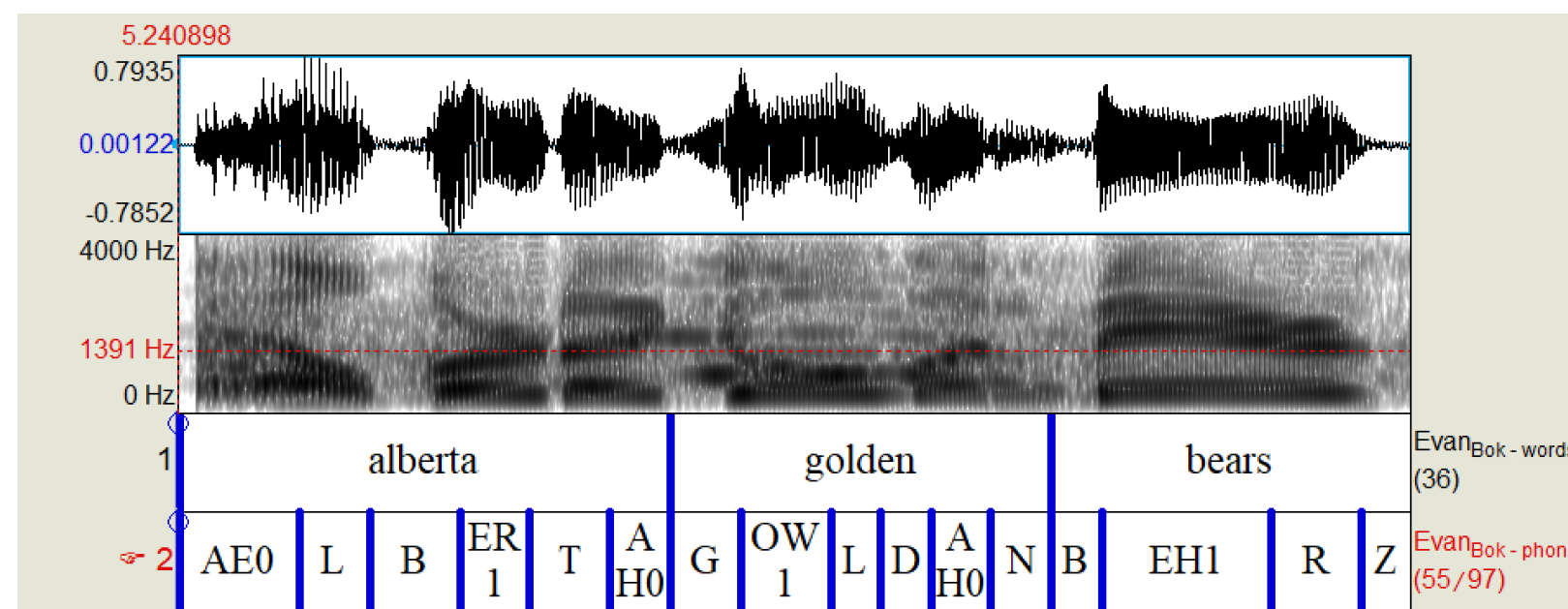
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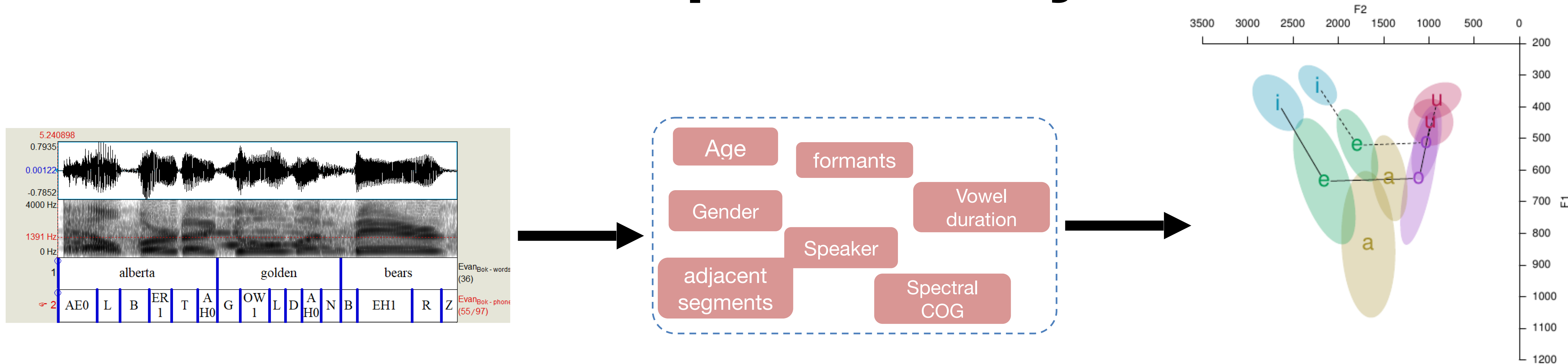
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2. Use AI model(s) to synthesise speech of those speakers
3. Segment/annotate both human and AI speech
4. Extract acoustic measure of interest
5. Statistically **compare** human and AI speech based on acoustic measures

Case Study

AI-generated English & Japanese stops

JASA
EXPRESS
LETTERS

ARTICLE

asa.scitation.org/journal/jel



Language-specific phonetic realisation of stop voicing contrasts in English and Japanese synthesised speech

James Tanner,^{1,a)}  Yasuaki Shinohara,²  and Faith Chiu¹ 

¹English Language and Linguistics, University of Glasgow, Glasgow, G12 8QQ, United Kingdom

²Faculty of Commerce, Waseda University, Tokyo, 169-8050, Japan

james.tanner@glasgow.ac.uk, y.shinohara@waseda.jp, faith.chiu@glasgow.ac.uk

Introduction

1. *Does the acoustic realisation of stop voicing in AI-generated speech follow the expected patterns from human speech?*
2. *When synthesising a speaker into a different language, does the speech follow the speaker's or language's stop voicing pattern?*

Introduction

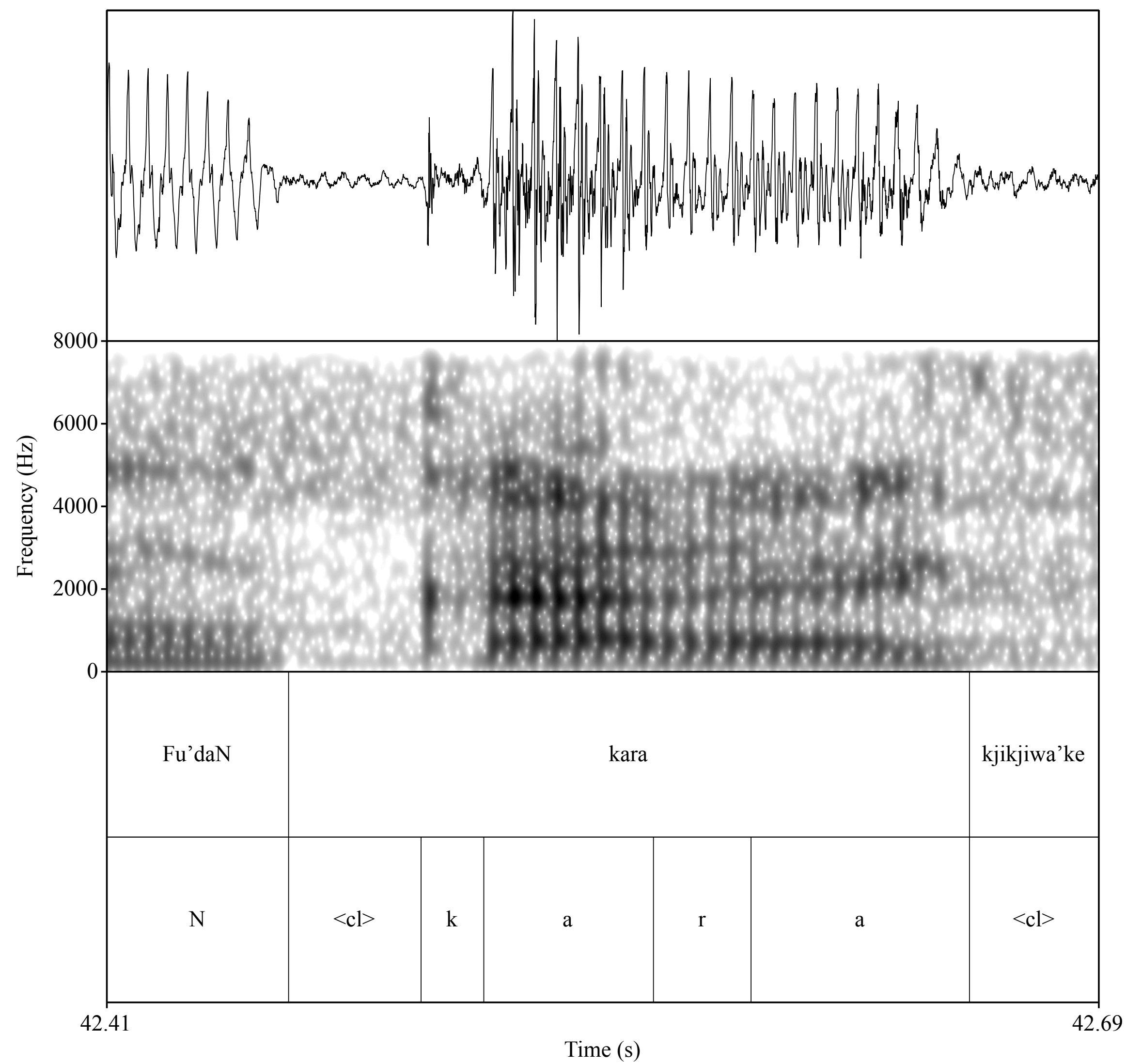
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- 2. When synthesising a speaker into a different language, does the speech follow the speaker's or language's stop voicing pattern?*

Here: examine acoustic variability in AI-generated Japanese and English stops with respect to 3 acoustic parameters:

- 1. Voice Onset Time (VOT)**
- 2. Closure Duration**
- 3. Closure Voicing**

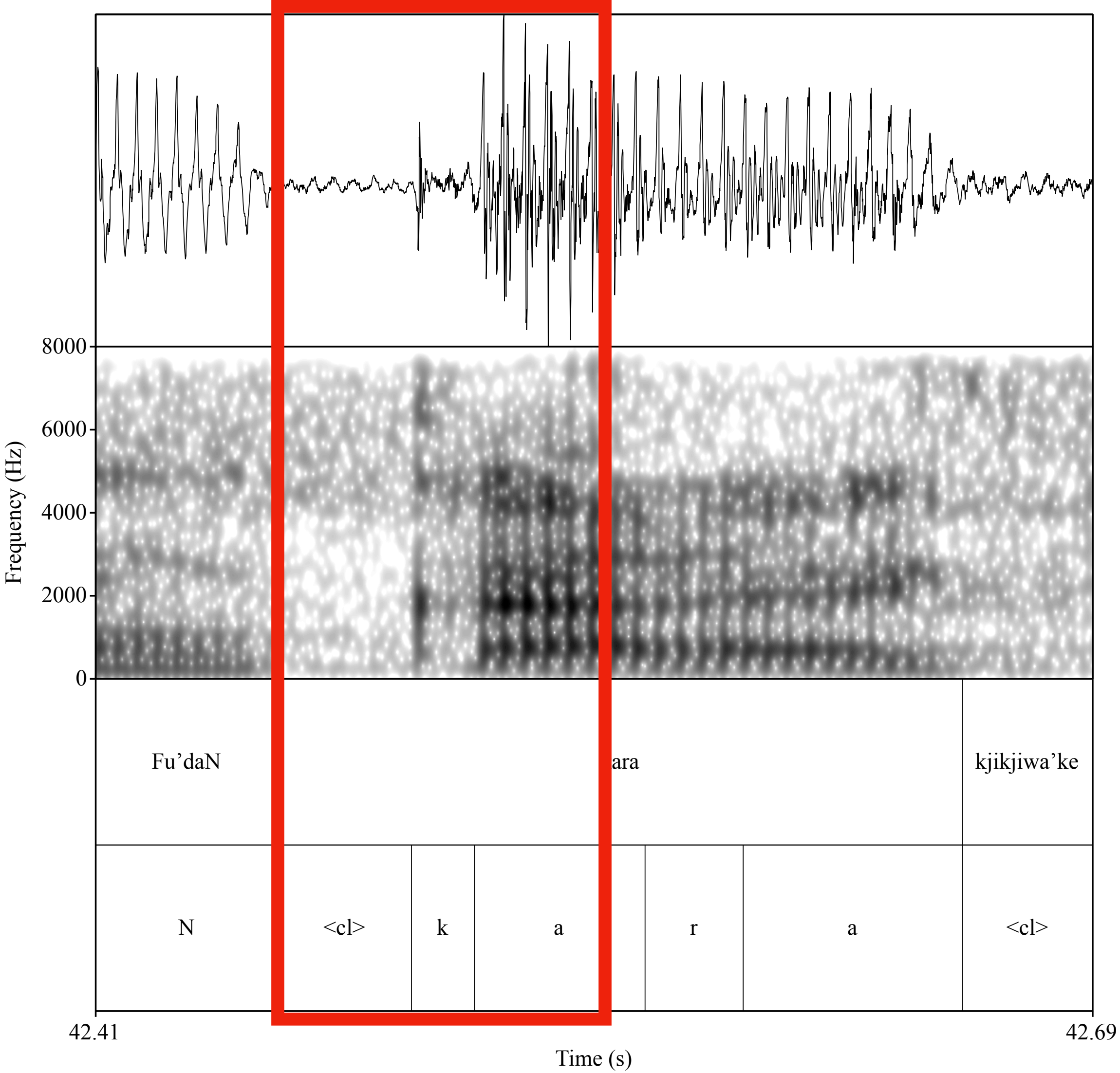
Phonetics of stops

- Stops involve a number of ordered phonetic events:



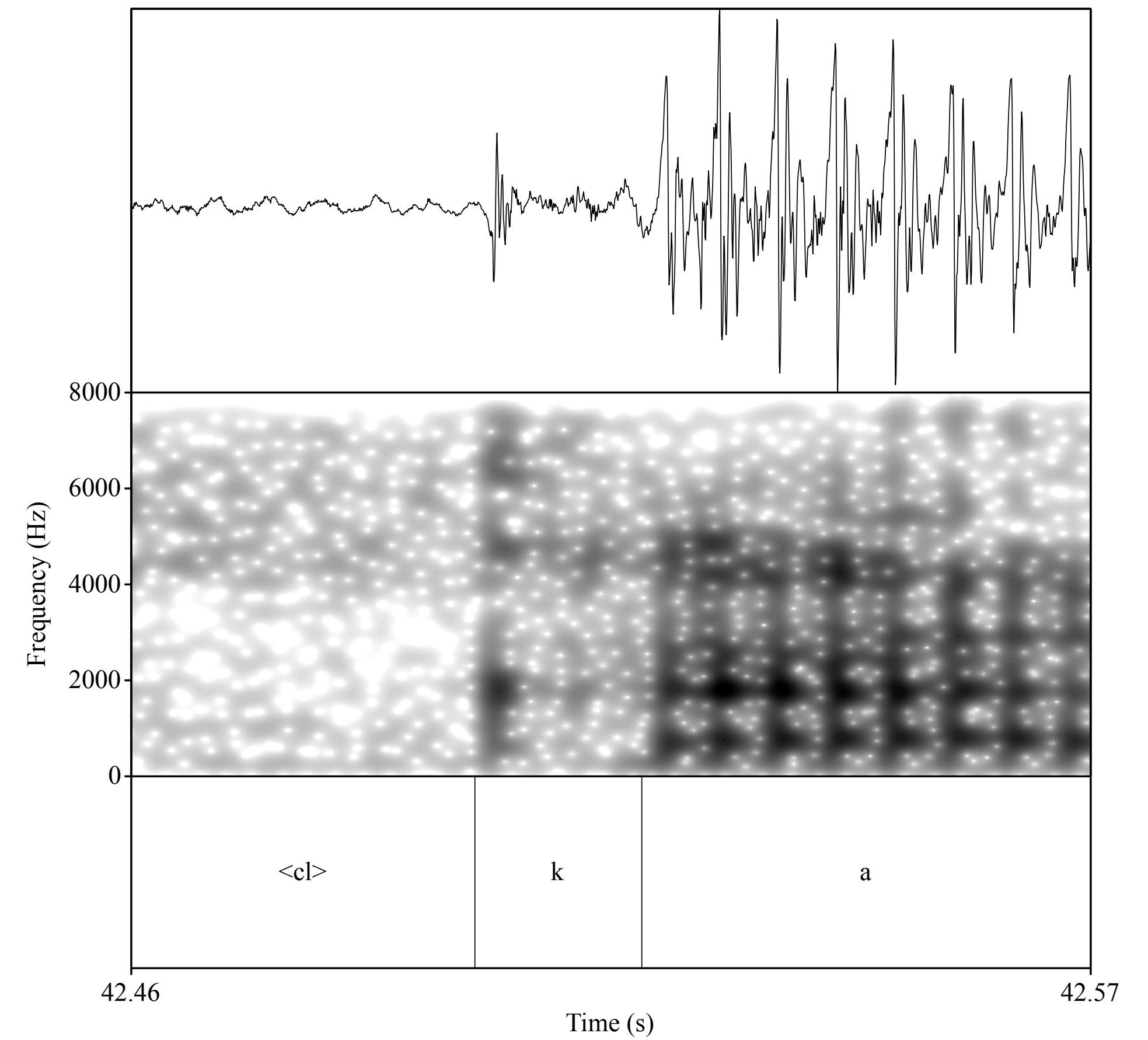
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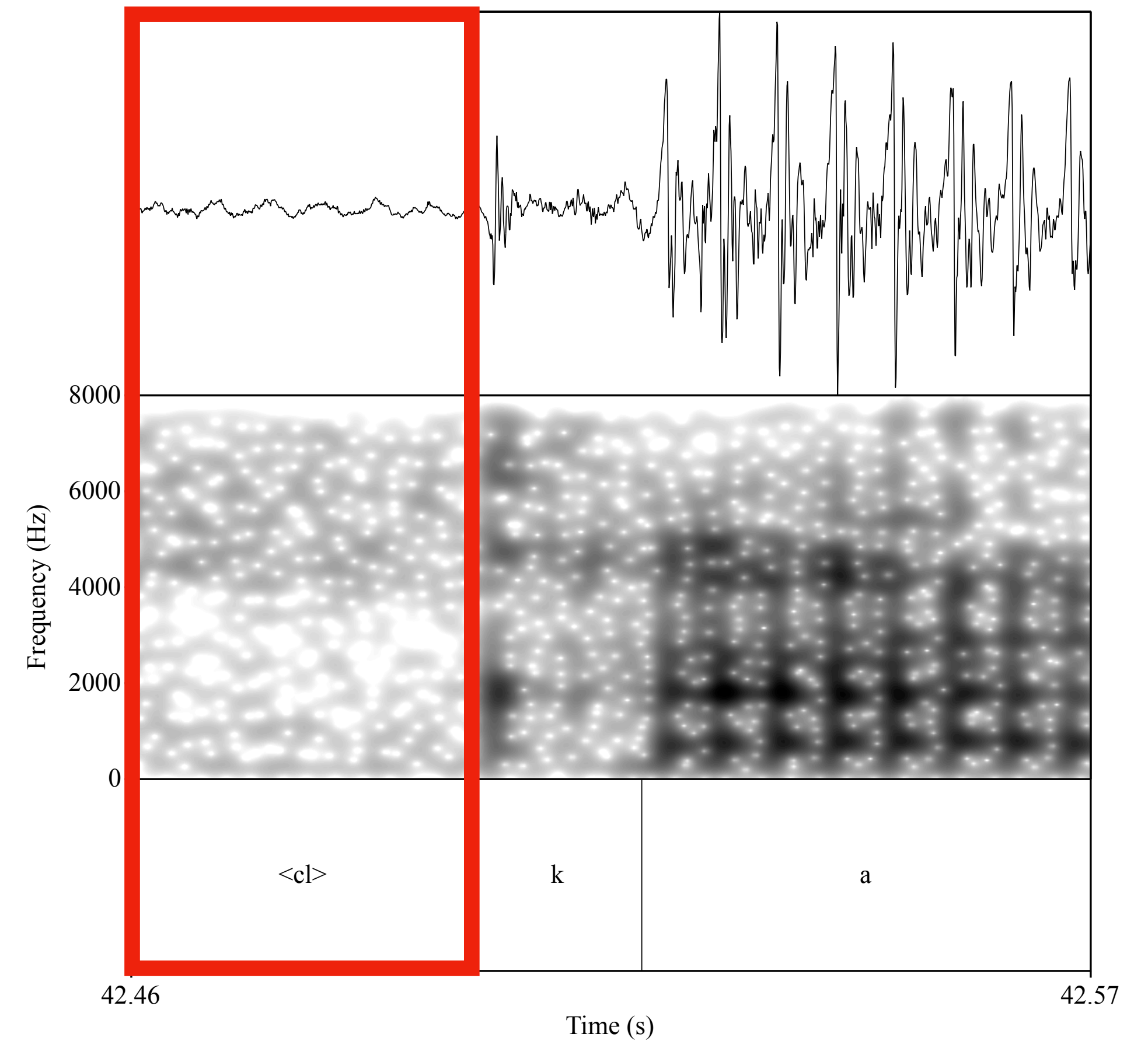
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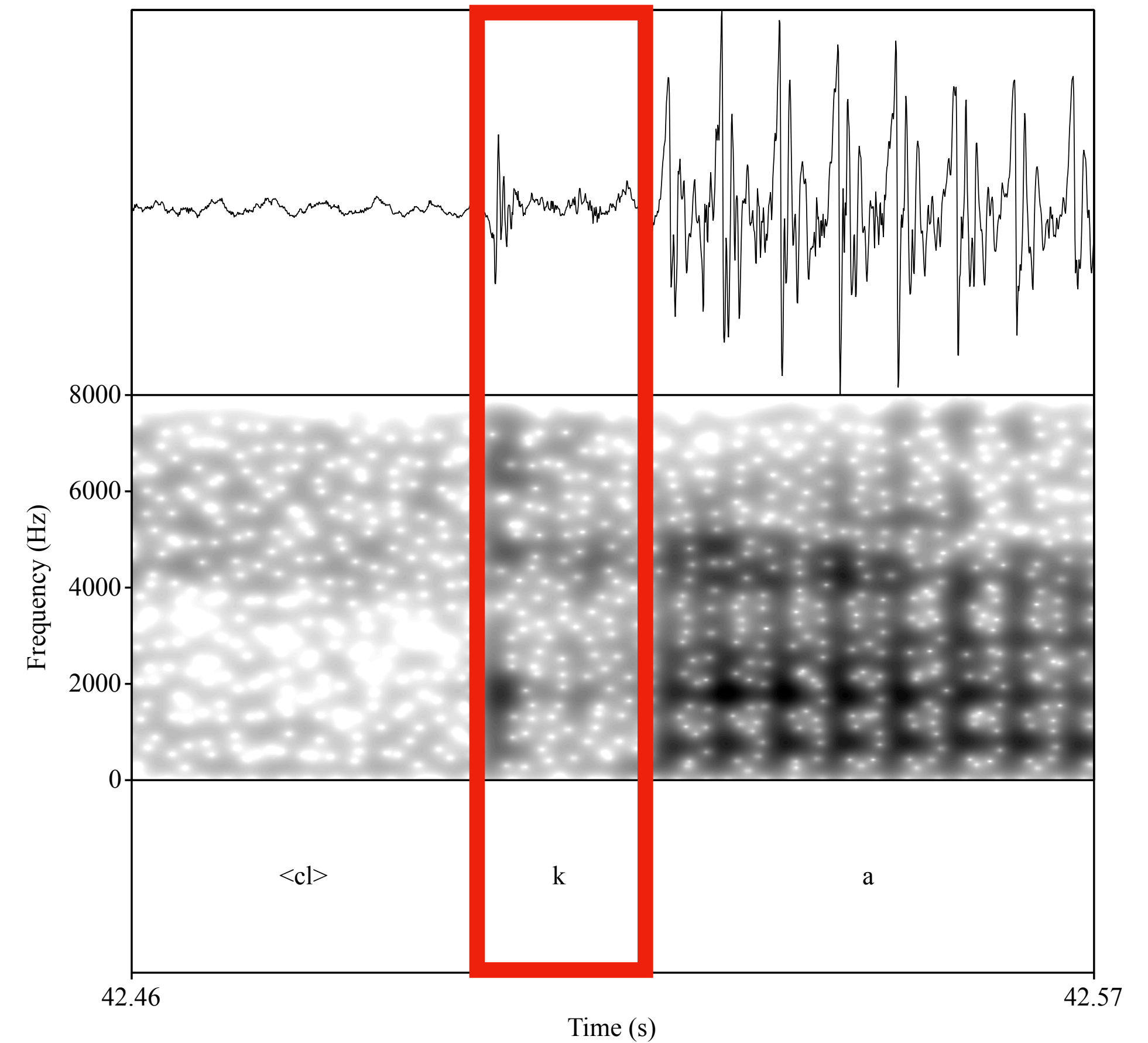
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 1. Closure of the oral cavity -> cessation of spectral activity



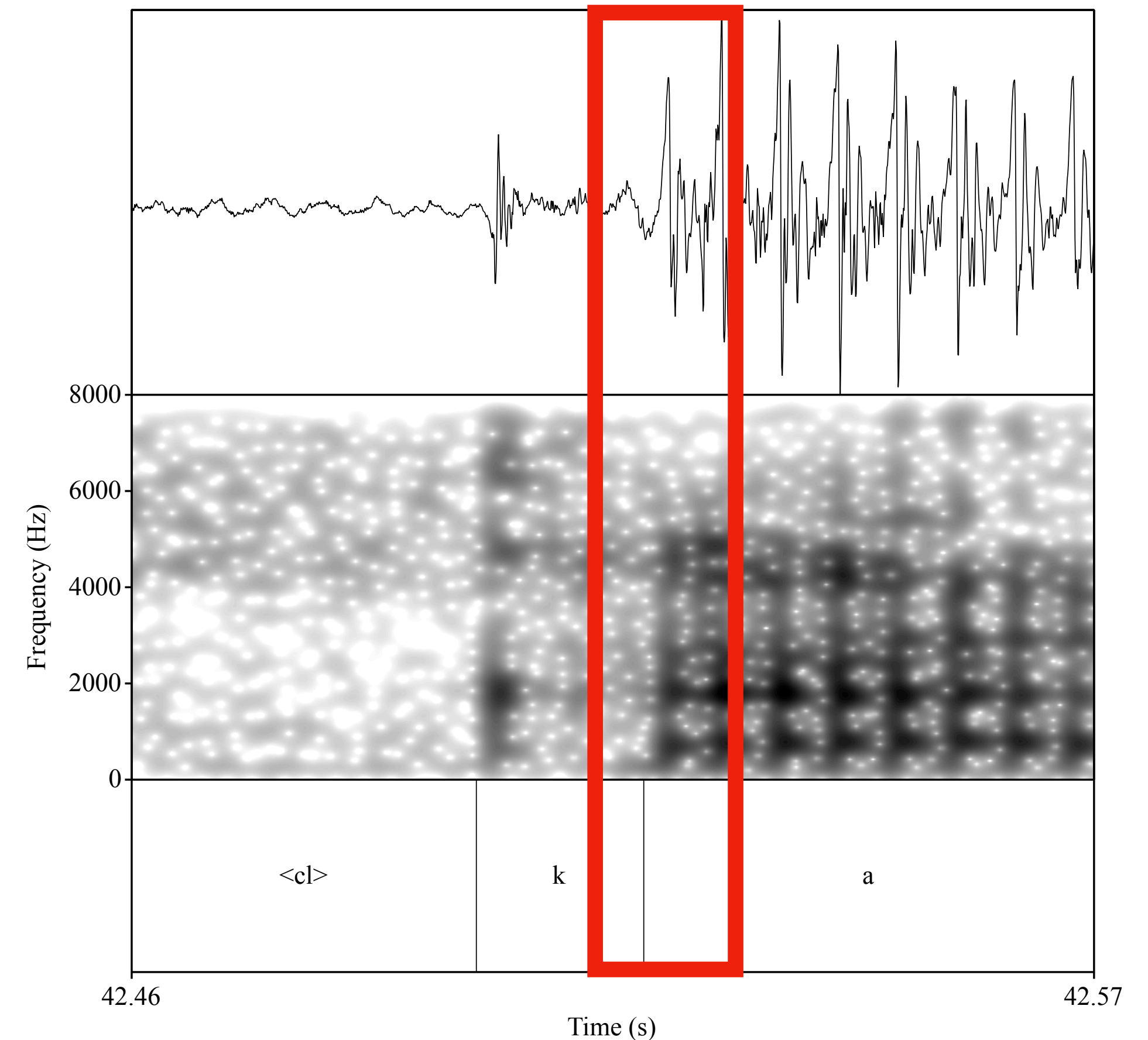
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 2. Oral closure release -> spike in energy



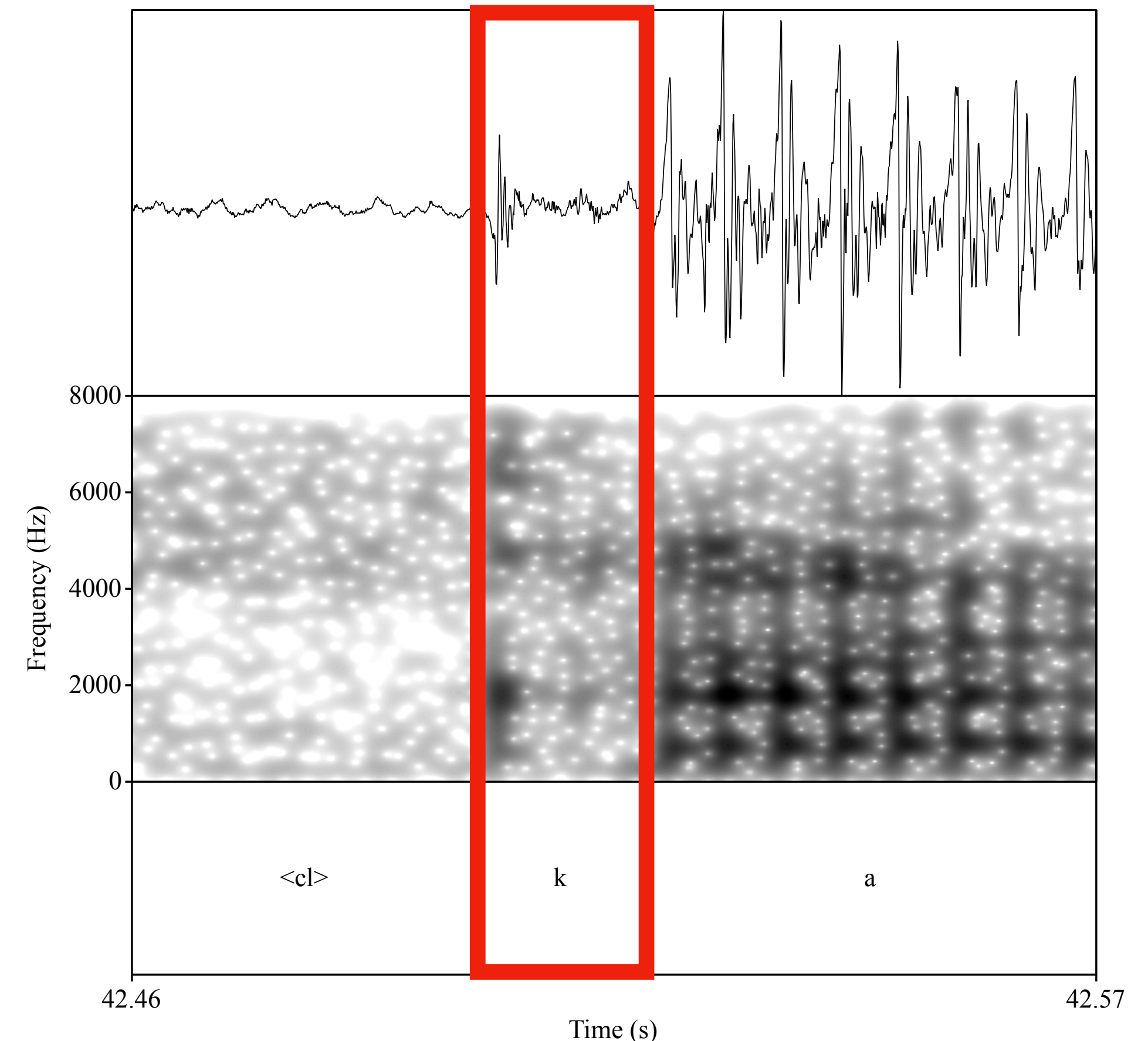
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Phonetics of stops

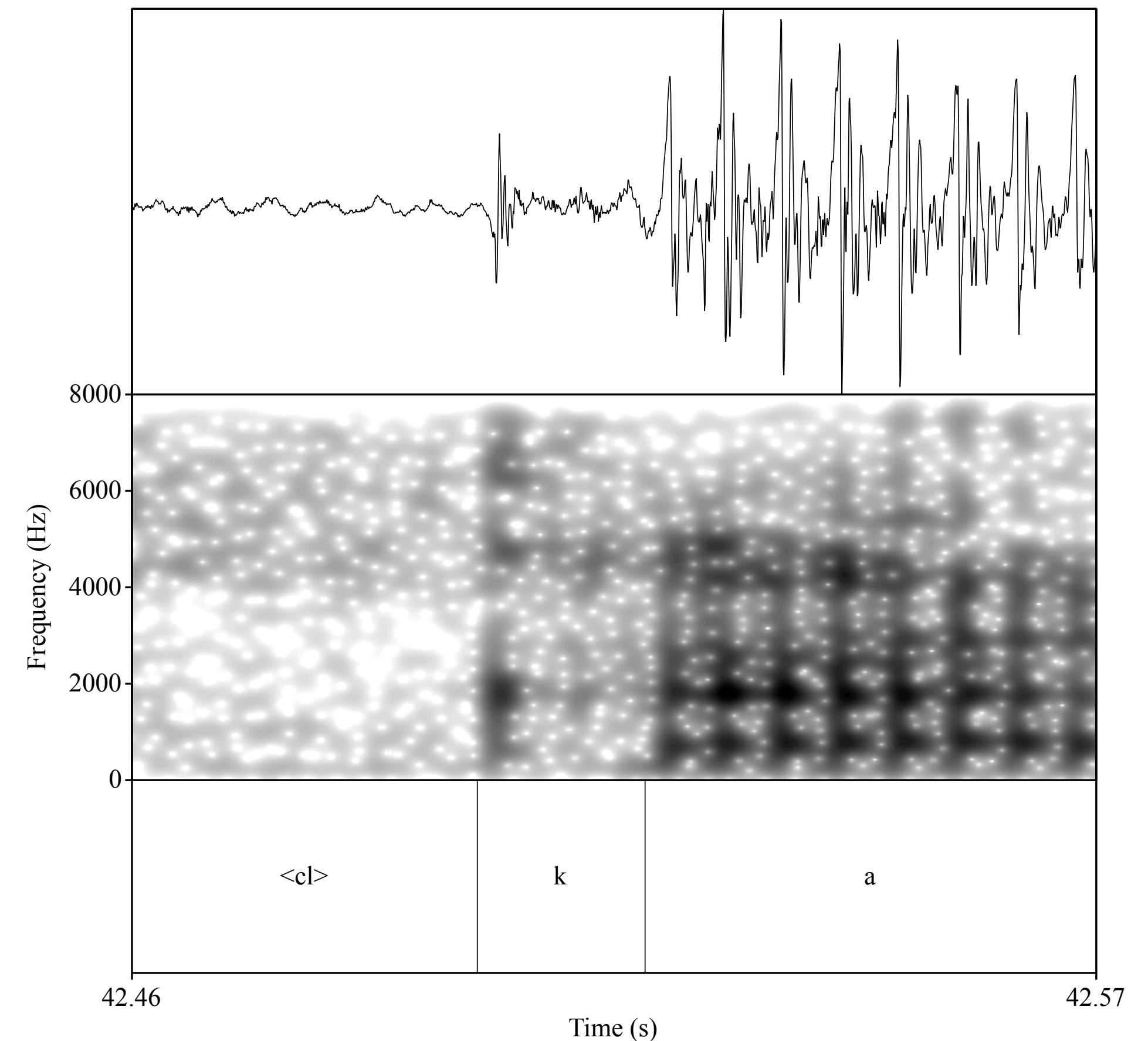
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Voice Onset Time (VOT) = time between closure release and beginning of voicing

Phonetics of stops

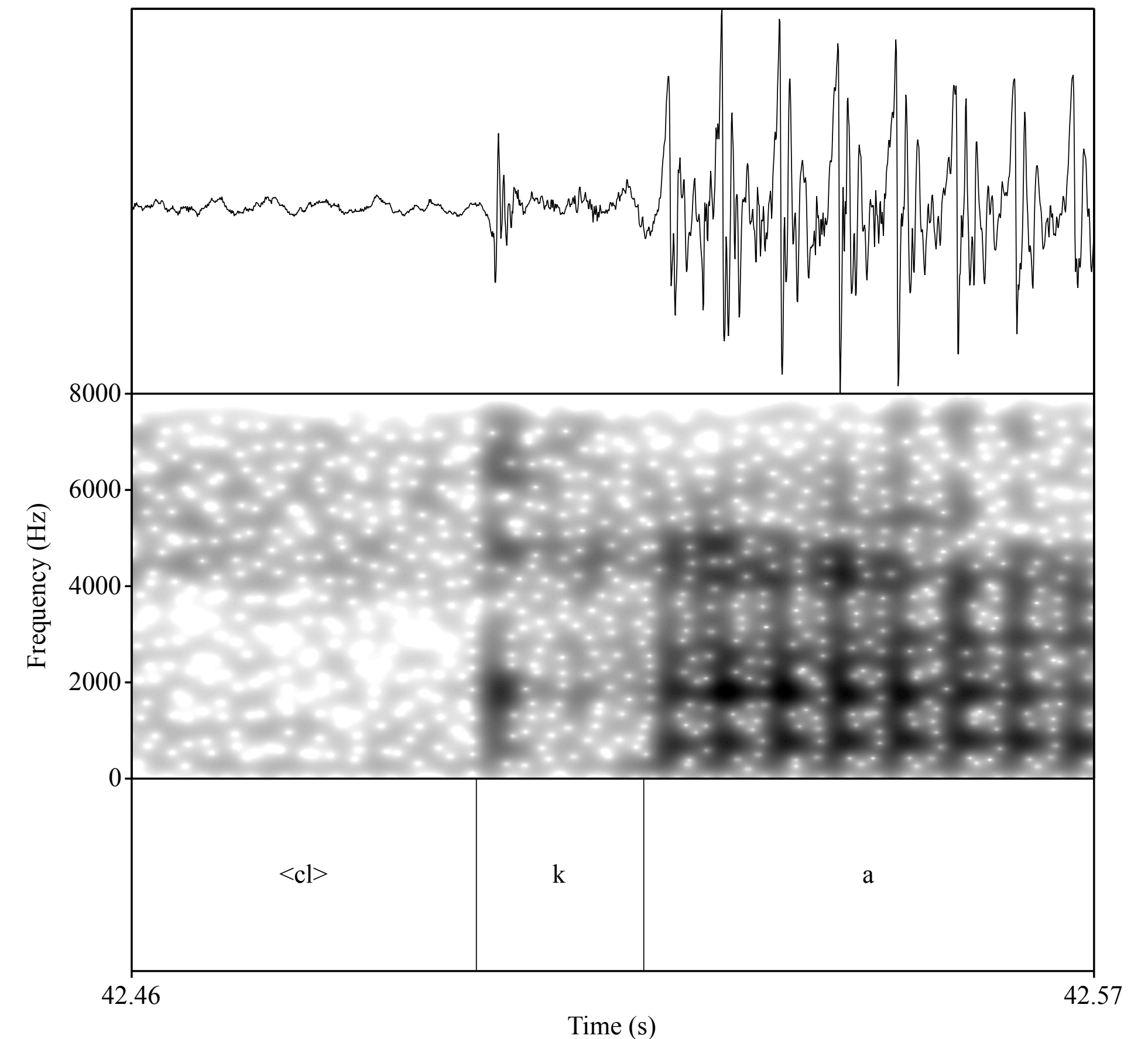
- Japanese stops characterised by:
 - Short VOT (12ms ~ 40ms)
 - Short Closure Duration (35ms ~ 65ms)
 - Controlled voicing during the stop closure
 - Voiceless stops < 10%
 - Voiced stops > 80%



Honma (1981), Shimizu (1996), Riney *et al.* (2007), Amano & Hirata (2010), Gao & Arai (2019), Tanner *et al.* (2020, 2025)

Phonetics of stops

- English stops characterised by:
 - Long VOT (20 ~ 60ms)
 - Long Closure Duration (50 ~ 80ms)
 - Closure voicing for both voiced and voiceless stops (20 ~ 70%)



Edwards (1981), Keating (1983), Docherty (1992), Byrd (1993) ,
Davidson (2016, 2017), Sonderegger *et al.* (2020)

Method: Data

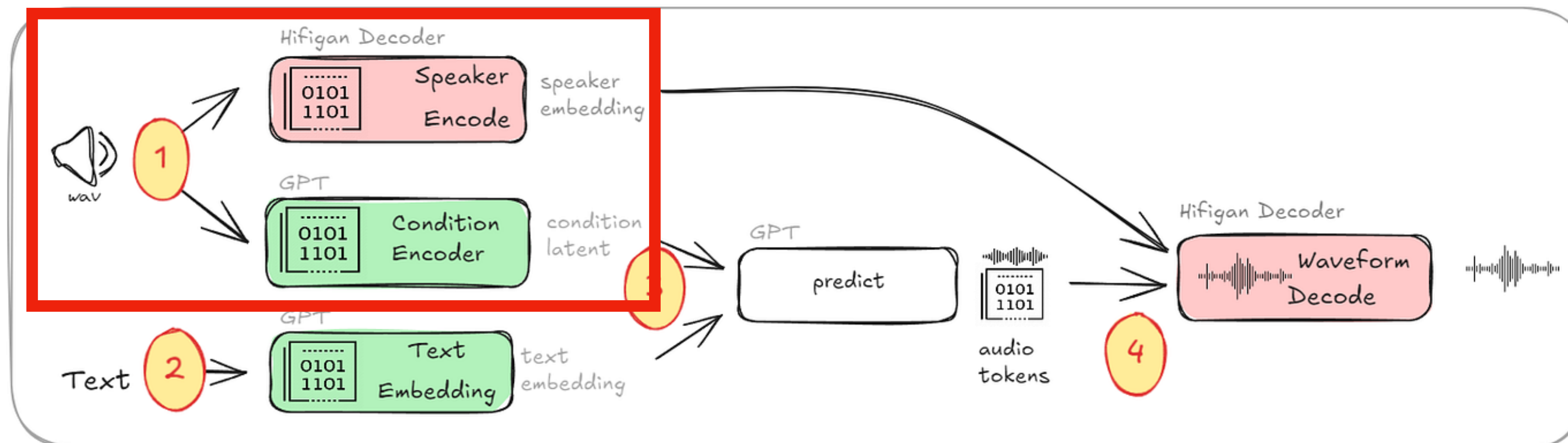
- Reference data from *University of Tsukuba Multi-Language Corpus*
 - Reading passage (*North Wind & Sun*) for 11 languages
 - Here: 8 English & 13 Japanese speakers

Method: Data

- Reference data from *University of Tsukuba Multi-Language Corpus*
 - Reading passage (*North Wind & Sun*) for 11 languages
 - Here: 8 English & 13 Japanese speakers
- Texts:
 - English: standard speech elicitation texts (582 sentences)
Rainbow Passage, Please Call Stella, Comma Gets a Cure, Arthur the Rat,
selected texts from TIMIT corpus
 - Japanese: emotionally-neutral texts from *Voice Actress Corpus* (100 sentences)

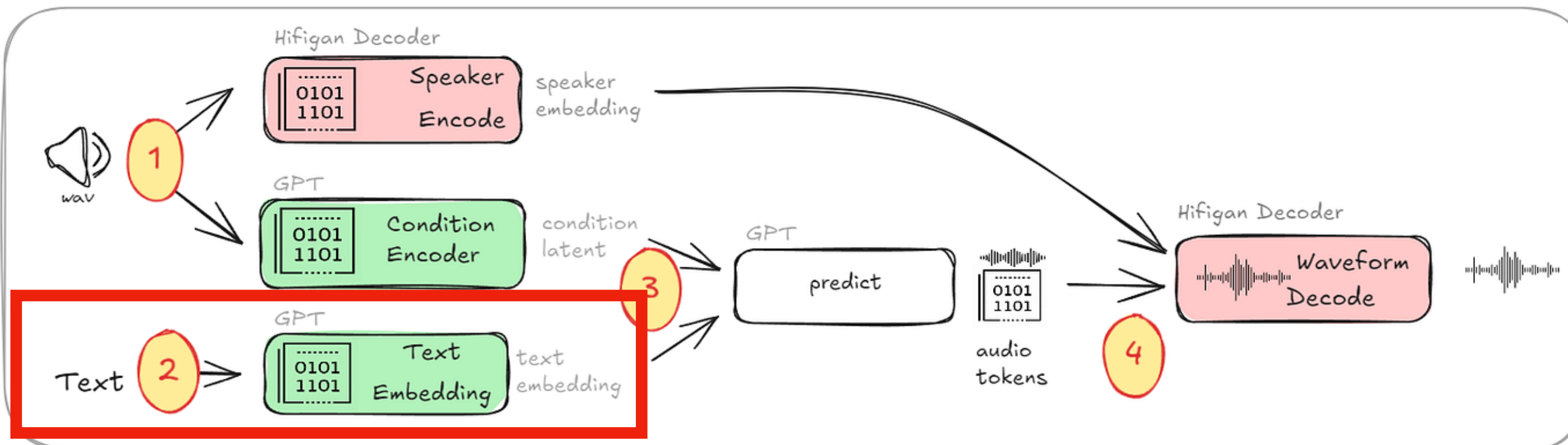
Method: Synthesis

- Speech samples synthesised using the *X-TTS* speech synthesis model
- For each speaker:
 - Reading passage audio passed to model to *condition* model on speaker
 -



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Synthesise speakers in both their *original* language (e.g. Japanese -> Japanese) and *opposite* language (e.g. Japanese -> English)

Examples

Within-language

Across-language

		Synthesised Language	
		English	Japanese
Speaker Language	English		
	Japanese		

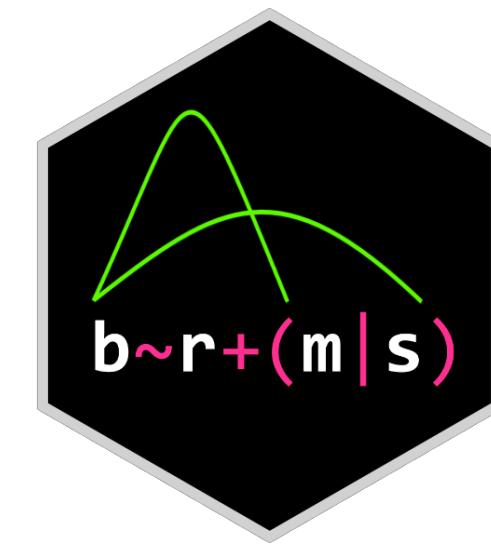
Method: Processing & Measurement

- Generated speech data aligned using *Montreal Forced Aligner*
- All utterance-medial stops extracted using *PolyglotDB*
- Acoustic measurement:
 - VOT estimated with *AutoVOT*
 - Closure duration: difference between start of aligned stop boundary and onset of VOT
 - Closure voicing: Praat *Fraction of Unvoiced Frames* within closure

Number of tokens per combination

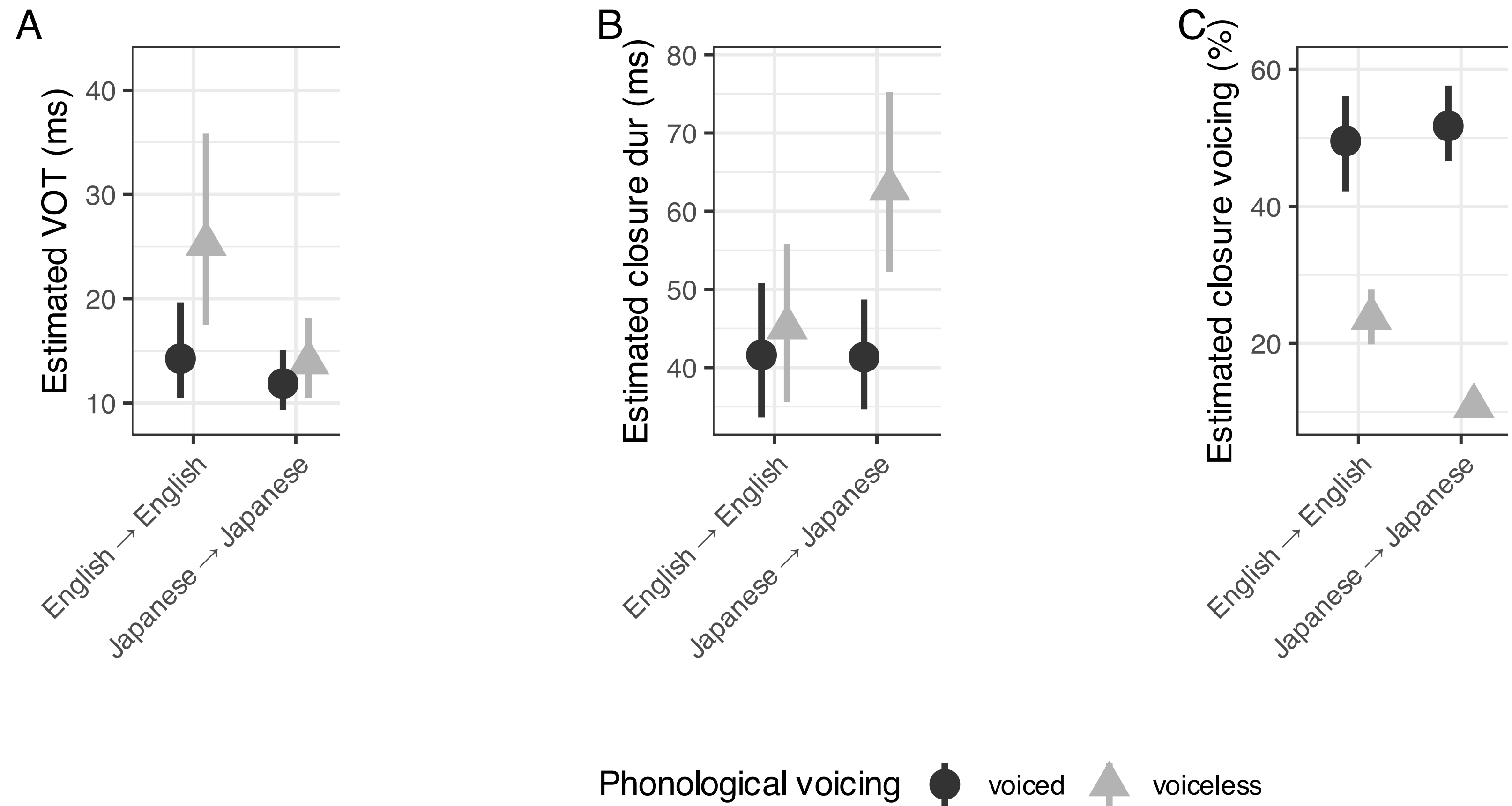
	Synthesised Language	
Speaker Language	English	Japanese
English	3672	2277
Japanese	6772	3911

Method: Statistical Analysis

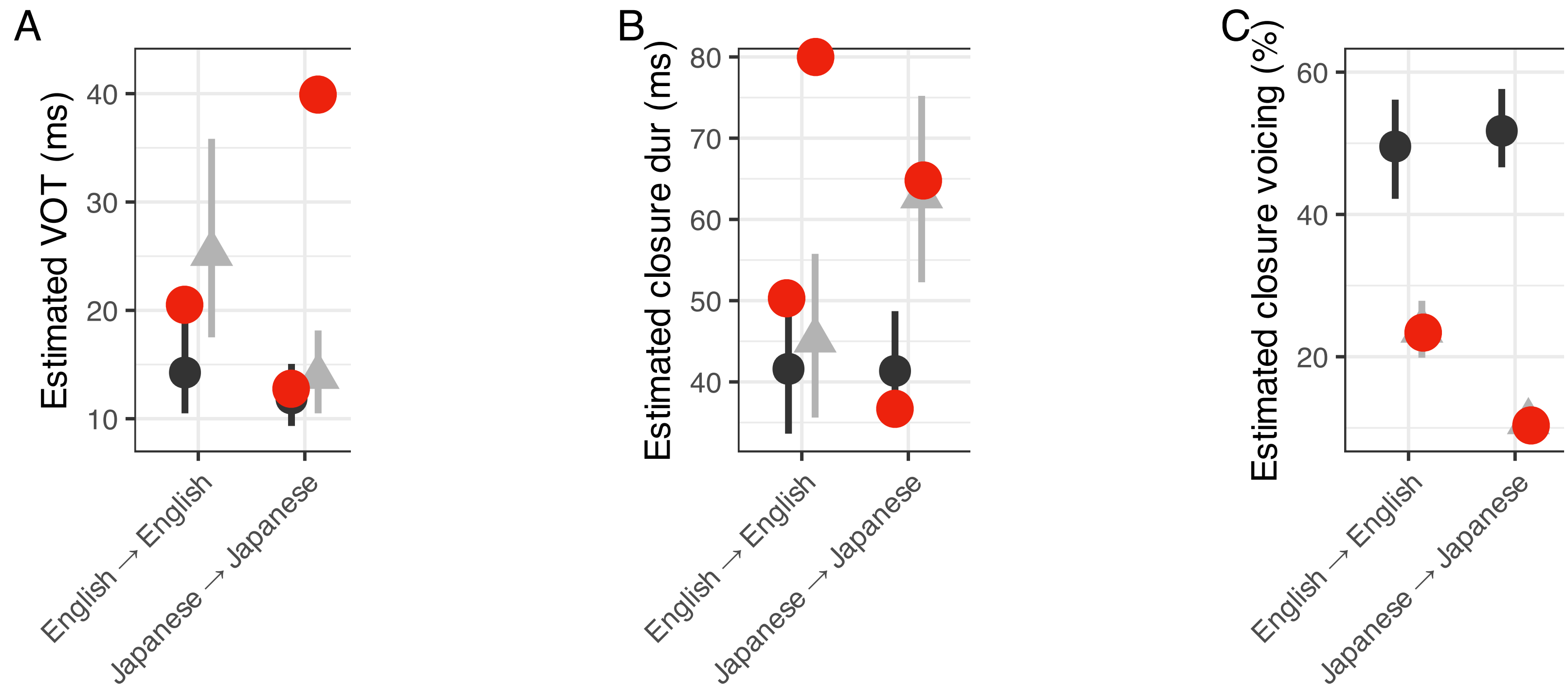


- Modelled with Bayesian regression analysis
- Speaker-Language/Synthesised-Language combination
{English-English, Japanese-Japanese, English-Japanese, Japanese-English}
- Stop voicing category {voiced, voiceless}
- Voicing - Combination interaction
- Controls:
 - Speech rate, place of articulation
- Random speaker-level terms

Results: same language



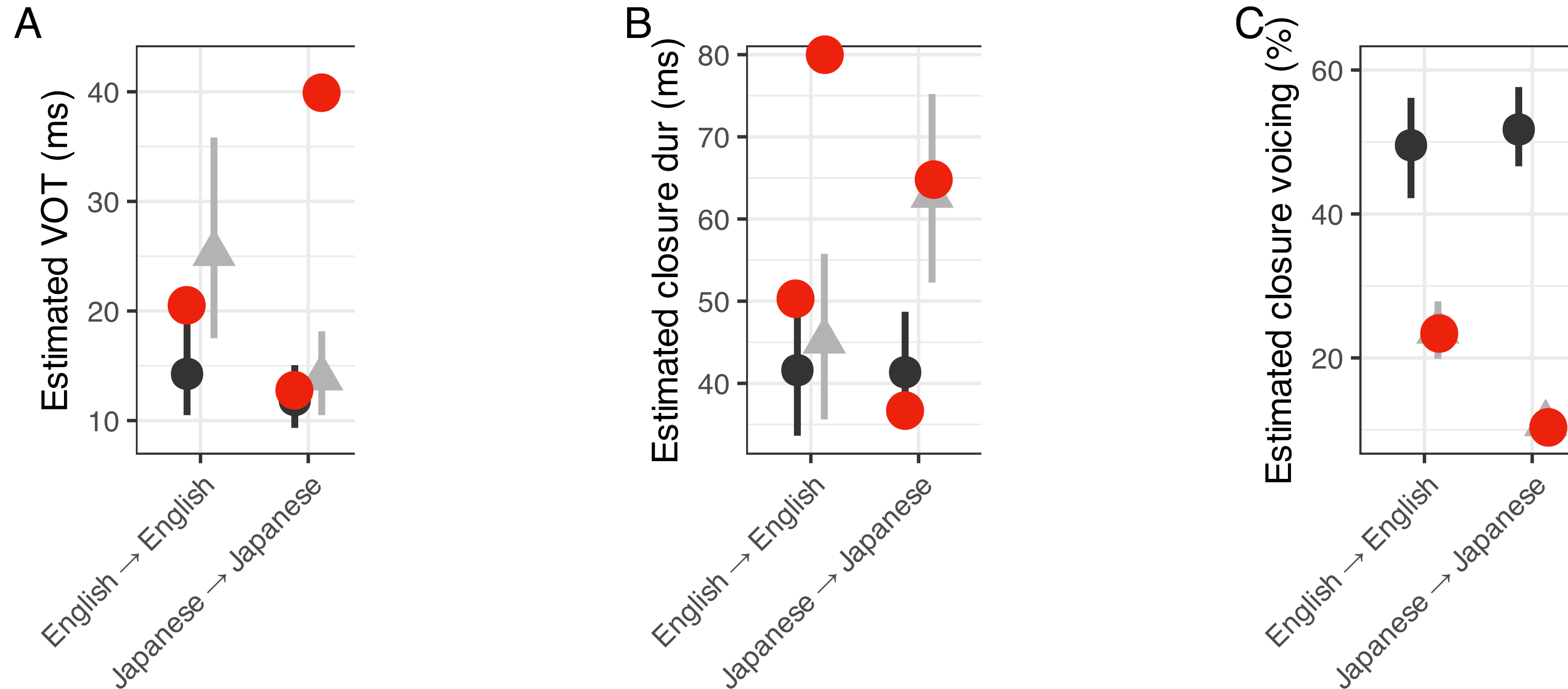
Results: same language



Expected value for human speech from literature

Phonological voicing ● voiced ▲ voiceless

Results: same language

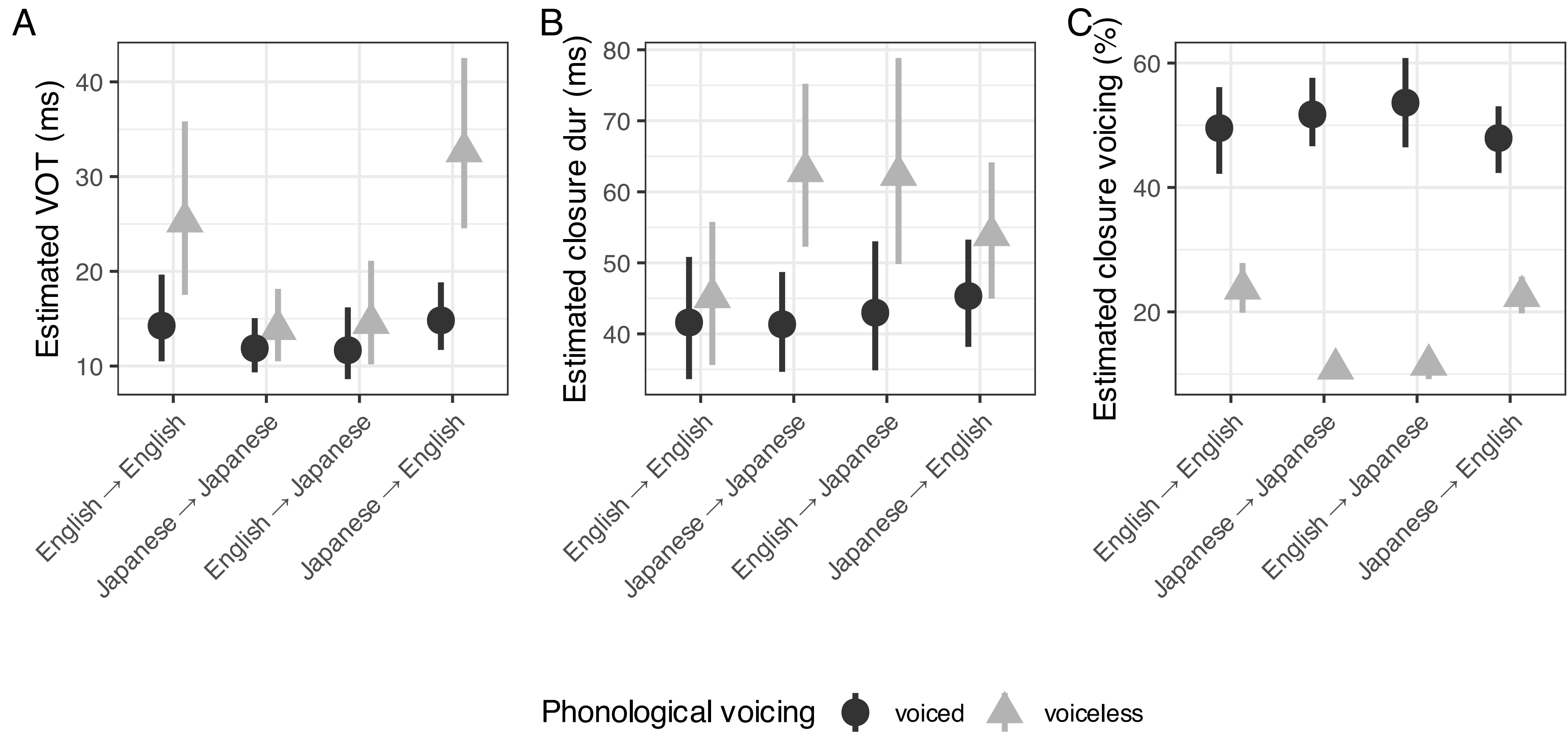


Expected value for human speech from literature

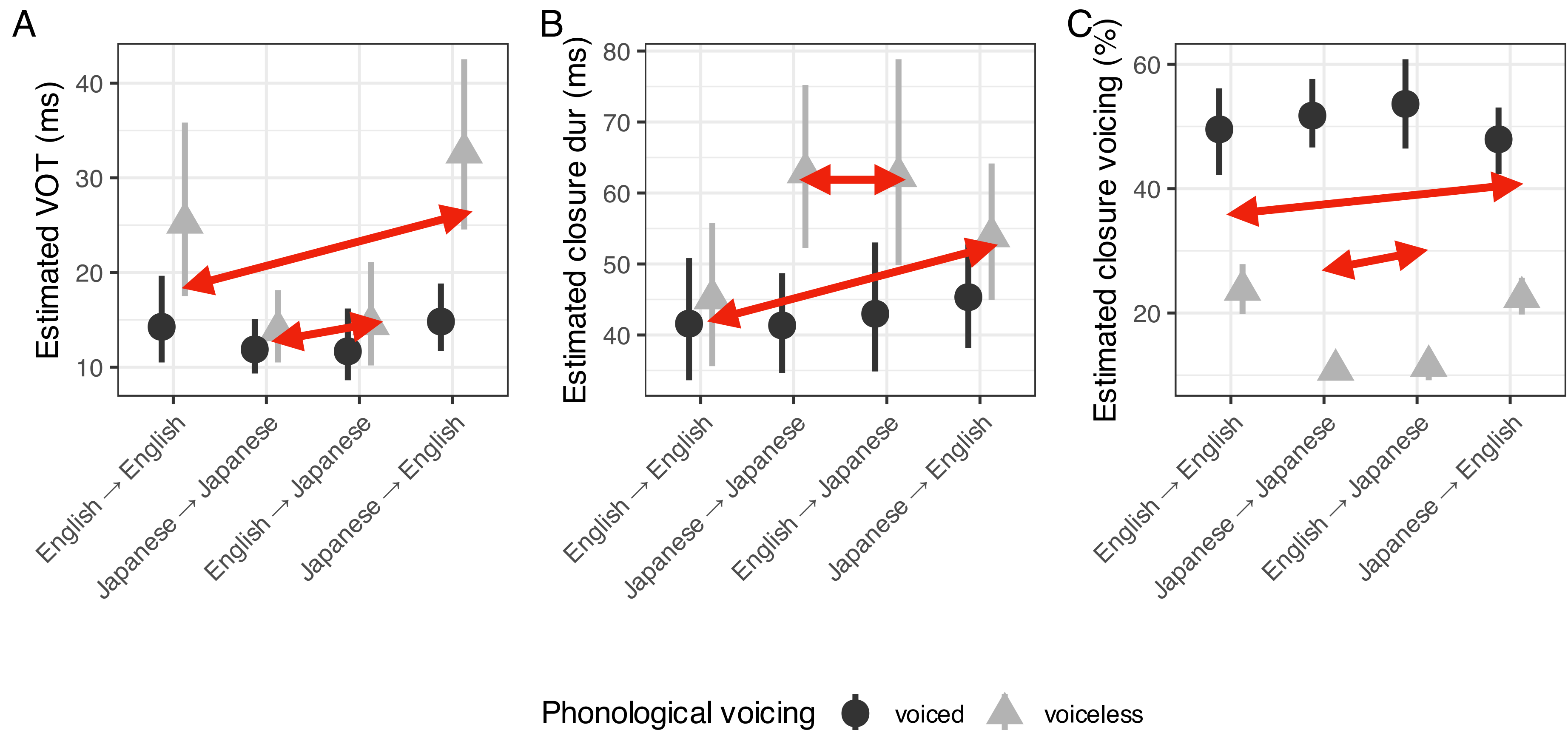
Phonological voicing ● voiced ▲ voiceless

AI-generated stops diverge quite dramatically from human stops!

Results: across language



Results: across language



Synthesised language > speaker's original language

Summary

1. *Does the acoustic realisation of stop voicing in AI-generated speech follow the expected patterns from human speech?*
2. *When synthesising a speaker into a different language, does the speech follow the speaker's or language's stop voicing pattern?*

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Acoustic realisation of AI-generated stops diverge from expectations from human speech literature

- **Voiceless VOT found to be substantially shorter for both Japanese and English**
- **Shorter closure duration for English**
- **Smaller degree of closure voicing**

Summary

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Stops synthesised in another language from the speaker (e.g., Japanese speaker into English) more closely pattern with the target language

Models are able to disentangle the language-specific and speaker-specific acoustic properties of stops

Future Directions

Future directions

- Case study examined ***just one*** type of segment (stops)
 - But observed large divergence from expected values (from previous lit)
- **Compare AI-generated phonetic properties *directly* with speech of the cloned speaker**
- Case study looked at outputs from one model (X-TTS)
 - *How much phonetic variability exists across AI speech models? Are some models more phonetically-faithful than others?*
- **Compare generated speech *across models***

Future directions

- More generally, case study shows that **phonetic methods can show differences between AI and human speech**
- Opens a wide range of possible research questions
 - *What kinds of speaker-specific phonetic variability are captured by AI models?*
 - *How do models generalise across speakers from a novel dialect?*
 - *What about child speech?*
- ***Can these acoustic-phonetic methods be used for building AI speech detection tools?***

Conclusion

- Modern AI systems have made it easier than ever to create high-quality speech of any speaker
 - Usually free & requiring minimal programming experience
- Applying acoustic-phonetic methods shows that AI-generated speech still differs from human speech
- Opens up the possibility to explore how linguistic knowledge is learned within AI models
- Also provides a route to AI speech detection using phonetically-intuitive information

Thank you!

Funding

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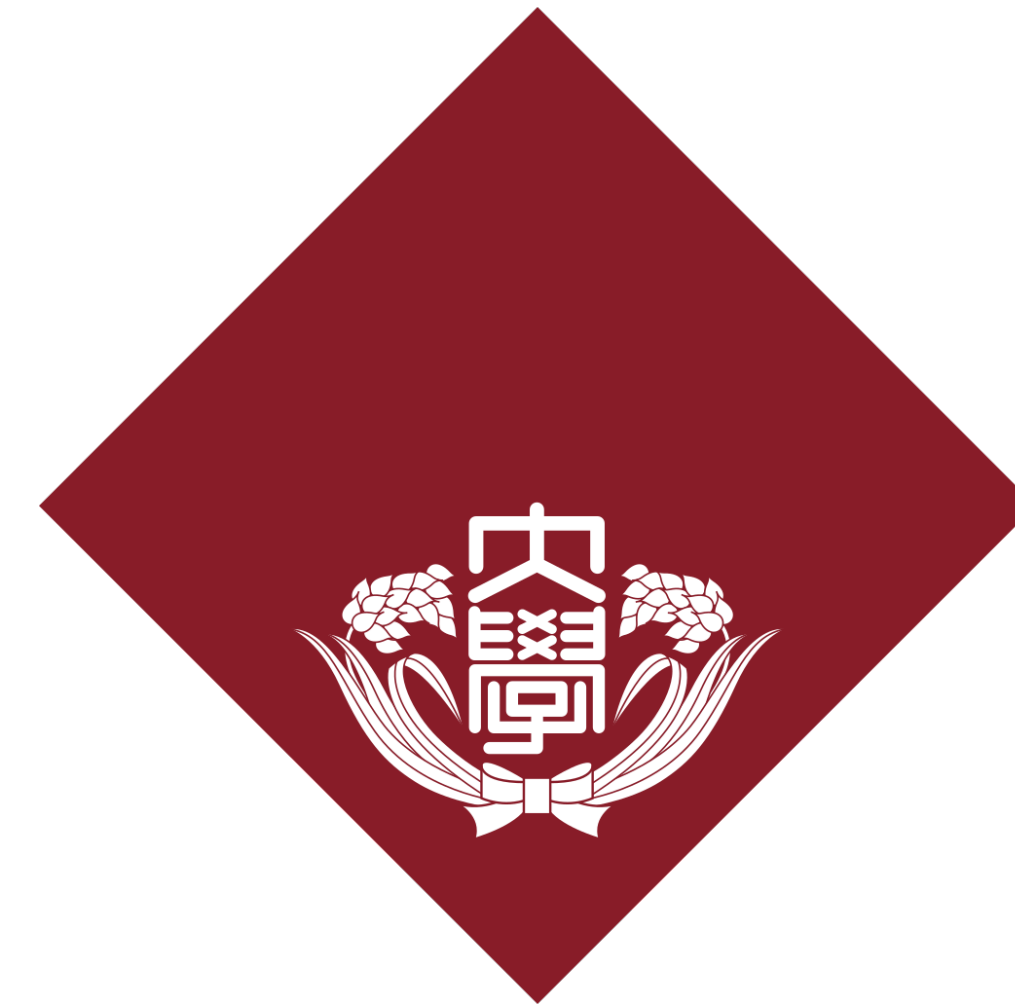
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Extra Slides

What would an AI speech study look like?

- Treat speech source (human vs AI) as a condition within an acoustic-phonetic study
- One (hypothetical) pipeline:
 1. Collect human speech data (or use collected corpus data)
 2. Use AI model(s) to synthesise speech of those speakers
 3. Segment/annotate both human and AI speech
 4. Extract acoustic measure of interest (e.g., vowel duration, vowel formants, stop duration, etc)
 5. Statistically **compare** human and AI speech

